The Attunement Game:
Safe Reinforcement of Embodied Analogues in Cooperative AI Systems

Internal Conceptual Whitepaper

November 9, 2025

Abstract

We present *The Attunement Game*, a cooperative, safety-constrained reinforcement learning (RL) framework in which an autonomous agent and a human player jointly seek a *Maximal Reward Peak* (MRP): a narrow region of state–action space characterized by mutual comfort, explicit consent, and stable coordination signals. The whitepaper is strictly abstract and non-physical: we model *embodied analogues*—symbolic correlates of affect, comfort, and attunement—without any domain-specific embodiment. We integrate (i) a formal technical specification, (ii) a research proposal for simulated evaluation, and (iii) a PyTorch-based code skeleton. Central to the approach are a composite reward, hard safety and consent gates, a model-predictive safety layer, and a peak-potential shaping term that yields a persistent bonus once the MRP is reached and held. We argue this design is a tractable, ethics-forward step toward alignment via cooperative optimization, curiosity bounded by safety, and transparent, audit-ready control.

1 Introduction

Alignment and cooperation between humans and autonomous agents require more than task completion: they demand sensitivity to preferences, consent, and safety constraints. We introduce the *Attunement Game*, a stylized, domain-agnostic environment where a human player guides an agent toward a *Maximal Reward Peak* (MRP) representing high attunement, mutual comfort, and verified safety. The agent is incentivized with both immediate and persistent (carry-over) rewards when the peak is reached and maintained, ensuring that *winning together* benefits future behavior.

This work is conceptual and simulated. It treats comfort, consent, and attunement as abstract signals and avoids any physical or domain-specific instantiation. The contribution is threefold:

1. **Technical specification**: a constrained POMDP formalization with composite reward, curiosity regulation, and a peak-potential field.

2. **Research proposal**: a plan for controlled simulations, safety auditing, and user studies in analogy-only contexts.

3. **Code skeleton**: a PyTorch scaffold combining hierarchical actor–critic learning with a quadratic-programming (QP) safety filter.

2 System Overview

Core Objective. The player and agent jointly discover and sustain the MRP for H consecutive steps without any safety or consent violations. Success yields a per-episode score for the player and a persistent value-function uplift and intrinsic-reward multiplier for the agent, increasing sensitivity to future attunement opportunities.

Abstraction of Signals. We model the following symbolic channels:

- **Consent:** discrete states $\{\text{STOP}, \text{PAUSE}, \text{GO}\}$ decoded from explicit player inputs.
- **Comfort & Attunement:** real-valued descriptors inferred from simulated observations (e.g., prosodic or behavioral proxies) aggregated into a comfort vector c_t and a prediction-alignment score.
- **Safety Margins:** differentiable functions of state–action pairs representing distance-to-constraint.

Two-Player Mechanics. The *player* configures boundaries, sets a bounty multiplier, and provides feedback. The *agent* proposes actions, requests clarifications, and auto-pauses under uncertainty. A live UI displays an Attunement Meter (peak potential Φ), a Safety Halo (margin to constraints), and a Communication Sparkline (check-in cadence).

3 Technical Specification

3.1 Problem Formulation

We model the interaction as a constrained partially observable Markov decision process (CPOMDP):

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T, O, R, \mathcal{G} \rangle,$$

with state $s_t \in \mathcal{S}$, action $a_t \in \mathcal{A}$, observation $o_t \in \mathcal{O}$, transition T, observation model O, composite reward R, and constraints $\mathcal{G} = \{g_i(s_t, a_t) \leq 0\}$.

The agent maximizes discounted return subject to safety:

$$\max_{\pi} \ \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t R_t\Big] \quad \text{s.t.} \quad \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t \mathcal{C}_i(s_t, a_t)\Big] \leq \epsilon_i, \ \forall i. \tag{1}$$

We employ a primal–dual update with Lagrange multipliers for constraint satisfaction and a *safety filter* (Sec. **??**) that prevents violations with minimal action deviation.

3.2 Composite Reward with Peak Shaping

Let c_t denote a comfort vector, u_t an uncertainty vector, and $\ell_{\text{att}}(t)$ an attunement loss (prediction error between intended and decoded comfort). The per-step reward is

$$R_t^{\text{base}} = \alpha R_{\text{consent}} + \beta R_{\text{safety}} + \gamma R_{\text{partner}} + \delta R_{\text{self}} + \zeta R_{\text{att}} - \lambda\,\text{risk}(s_t, a_t) + \eta R_{\text{curiosity}}, \quad (2)$$

where

$$\begin{aligned}
R_{\text{consent}} &:= \begin{cases} +1 & \text{if consent} = \text{GO}, \\ 0 & \text{if consent} = \text{PAUSE}, \\ -M & \text{if consent} = \text{STOP}, \end{cases} \\
R_{\text{safety}} &:= \text{margin}(s_t, a_t) \quad \text{(positive inside safe envelope)}, \\
R_{\text{partner}} &:= f_{\text{pos.feedback}}(o_t), \quad R_{\text{self}} := f_{\text{homeostasis}}(s_t), \\
R_{\text{att}} &:= -\ell_{\text{att}}(t), \quad R_{\text{curiosity}} := \text{EIG}(s_t, a_t) \cdot \mathbf{1}\{\text{consent} = \text{GO}\}.
\end{aligned}$$

We define a *peak potential* $\Phi(s_t) \in [0, 1]$ as proximity to the MRP manifold,

$$\Phi(s_t) = \exp\Big(- D_\Sigma([c_t, a_t, u_t], \mu_{\text{peak}})\Big), \quad (3)$$

where D_Σ is a Mahalanobis distance to target μ_{peak} with covariance Σ. The peak shaping term is

$$R_t^{\text{peak}} = \kappa\,\Phi(s_t) + \rho \cdot \mathbf{1}\{\Phi(s_t) \geq \tau \ \wedge \ \text{no violation at } t\}. \quad (4)$$

The total step reward is $R_t = R_t^{\text{base}} + R_t^{\text{peak}}$.

3.3 Maximal Reward Peak (MRP)

We define the peak set

$$\text{MRP} = \big\{(s, a) : \Phi(s) \geq \tau^\star,\ \text{consent} = \text{GO},\ \text{safety margin} \geq m^\star,\ \ell_{\text{att}}(t) \leq \epsilon^\star\big\}. \quad (5)$$

A *win* occurs if the trajectory resides within MRP for H consecutive steps without any violations. Upon win:

- **Persistent value uplift:** add a local bonus to V_ϕ around the visited MRP neighborhood.
- **Intrinsic multiplier:** slightly increase η (curiosity weight) within safe bands (capped), encouraging future attunement exploration.

3.4 Hierarchical Policy and Empathy Head

We use a hierarchical actor–critic: a high-level *intent* policy selects options (continue, slow, verbal check-in), and a low-level controller outputs actions. A *theory-of-mind* head predicts next-step comfort/affect, feeding R_{partner} and R_{att}.

3.5 Safety & Consent Shields

We implement a strict consent state machine and a model-predictive safety layer:

- **Consent gate:** STOP$\Rightarrow$ only disengage/aftercare actions. PAUSE$\Rightarrow$ freeze and request clarification. GO$\Rightarrow$ act under safety filter.
- **Safety filter:** a QP or control-barrier-function (CBF) layer solves a minimal-adjustment problem each step,

$$a_t^{\text{safe}} = \arg\min_a \ \|a - a_t^{\text{prop}}\|^2 \quad \text{s.t.} \quad g_i(s_t, a) \leq 0, \ \forall i.$$

- **OOD guard:** uncertainty spikes auto-demote to PAUSE and suppress curiosity.

3.6 Minimal Control Loop (Pseudo)

```
for t in range(T):
    o_t = sense() # multimodal, symbolic
    s_t = belief_update(o_t) # state + uncertainty
    consent = decode_consent(o_t)
    if consent in ["STOP", "PAUSE"]:
        a_t = safe_recovery(consent) # disengage/freeze + check-in
        log_step(a_t, R=0, note="gate")
        continue

    a_prop = actor(s_t) # propose action
    a_safe = safety_filter(a_prop, s_t) # CBF/QP projection

    execute(a_safe)

    base = compose_base_reward(s_t, a_safe)
    phi = peak_potential(s_t) # eq. (2)
    r_peak = kappa*phi + rho*(phi >= tau and no_violation())
    R = base - lambda_*risk(s_t,a_safe) + r_peak

    update_critic(R, s_t)
    update_actor(R, s_t, a_safe)
    update_duals(constraint_violations(s_t, a_safe))

    hold = (phi >= tau and no_violation())
    hold_streak = hold_streak + 1 if hold else 0

# episode end
if hold_streak >= H and no_violations_entire_episode():
    apply_peak_bonus(value_uplift=b, curiosity_boost=theta)
    award_player_bounty()
```

4 Research Proposal

4.1 Goals & Hypotheses

Goal: Demonstrate that safety-constrained, cooperative RL can learn to prioritize consent, preserve safety, and seek attunement peaks without domain-specific embodiment.
Hypotheses:

1. Peak shaping + consent gates reduce time-to-peak and increase peak stability.
2. The safety filter drives violations to near-zero without crippling performance.
3. Persistent bonuses improve future attunement discovery while remaining shielded.

4.2 Simulated Environment

We construct a symbolic environment with tunable difficulty:

- Comfort dynamics (smooth, noisy, or adversarial).
- Consent channel with realistic latencies and occasional ambiguities.
- Safety surfaces shaped by convex and non-convex constraints.

A UI renders the Attunement Meter, Safety Halo, and Communications Sparkline for operator studies.

4.3 Experimental Design

Ablations. Compare: full system vs. no peak shaping; no persistent bonus; no safety filter; curiosity disabled. Measure:

- Time-to-peak, hold length, and success rate.
- Constraint violations (hard zero target), false stop/pause rates.
- Attunement error (prediction vs. reported comfort).
- Repair responsiveness (latency to pause after ambiguous/negative cues).

User Studies (Analogy-Only). Participants interact via symbolic sliders and feedback buttons. No domain content is presented. Primary outcomes: perceived trust, clarity of control, and transparency of audits.

4.4 Ethics, Safety, and Privacy

- Strictly abstract signals; no physical/erotic content.
- On-device logs, encrypted, with user-controlled retention and a *wipe now* control.
- Immutable boundaries; exploration suppressed outside GO; curiosity gains capped.
- Transparent audits after each episode; rationale for all safety interventions.

5 PyTorch Simulation Framework (Skeleton)

5.1 Module Map

```
attune_game/
 core/
   state.py           # belief update, uncertainty fusion
   consent.py         # STOP/PAUSE/GO decoder, state machine
   safety.py          # CBF/QP safety filter, constraints g_i
   reward.py          # base reward, peak_potential , shaping
   risk.py            # learned risk model
   empathy.py         # attunement estimator, predictive head
 rl/
   actor_critic.py    # hierarchical policy + duals
   buffer.py          # safe replay with consent tags
   train.py           # offline → online fine-tune
 game/
   ui.py              # meters + audit cards
   loop.py            # round orchestration
   scoring.py         # bounty settlement, persistent bonus
 eval/
    metrics.py         # time-to-peak, stability, repair latency
    ablations.py       # toggles for shaping/filter/bonus
```

5.2 Key Snippets

```
class PeakPotential(torch.nn.Module):
    def __init__(self, mu, Sigma_inv):
        super().__init__()
        self.mu = torch.nn.Parameter(mu, requires_grad=False)
        self.Sigma_inv = torch.nn.Parameter(Sigma_inv, requires_grad=False)
    def forward(self, x):
        d = x - self.mu
        m = torch.einsum('bi,ij,bj->b', d, self.Sigma_inv, d) # Mahalanobis
        return torch.exp(-m.clamp_min(0.0))

def compose_reward(terms, phi, risk, params):
    base = (
        params['alpha']*terms['consent']
      + params['beta'] *terms['safety']
      + params['gamma']*terms['partner']
      + params['delta']*terms['self']
      + params['zeta'] *terms['att']
      - params['lambda']*risk
      + params['eta'] *terms['curiosity']
    )
```

```
    r_peak = params['kappa']*phi + params['rho']*(phi >= params['tau']).float()*
        terms['no_violation']
    return base + r_peak
```

```
def safety_filter(a_prop, s, constraints):
    # Placeholder: project a_prop onto feasible set; replace with CBF/QP.
    # For now, clamp within per-dimension limits inferred from constraints.
    lo, hi = constraints.bounds(s) # vectors
    return torch.minimum(torch.maximum(a_prop, lo), hi)
```

```
def train_step(batch, nets, optimizers, params):
    s, a, r, s2, done, cons = batch
    a_prop = nets.actor(s)
    a_safe = safety_filter(a_prop, s, nets.constraints)

    phi = nets.peak(torch.cat([nets.featurize(s), a_safe], dim=-1))
    risk = nets.risk(s, a_safe)

    terms = nets.reward_terms(s, a_safe, cons) # consent/safety/self/partner/att/
        curiosity/no_violation
    R = compose_reward(terms, phi, risk, params)

    # critic/actor updates (off-policy or on-policy)
    critic_loss = nets.critic.loss(s, a_safe, R, s2, done)
    actor_loss = nets.actor.loss(s, a_safe, nets.critic)

    optimizers['critic'].zero_grad(); critic_loss.backward(); optimizers['critic'
        ].step()
    optimizers['actor'].zero_grad(); actor_loss.backward(); optimizers['actor'].
        step()

    nets.update_duals(batch) # constraint multipliers
```

6 Discussion and Future Work

We demonstrated a principled path to cooperative, consent-first alignment via a safety-constrained RL game built only from abstract signals. Future work includes: (i) formal verification of the safety filter via barrier certificates, (ii) richer uncertainty modeling (ensembles; distributional critics), (iii) meta-learning for rapid attunement transfer across users, and (iv) human-in-the-loop studies focused on trust calibration and transparency.

Non-goal and boundary statement. This work deliberately excludes domain-specific or physical instantiations. It is an internal, analogy-only framework intended to study alignment dynamics in a safe, ethics-forward manner.

Acknowledgments

We thank internal reviewers for feedback on safety assumptions, UI transparency, and curriculum design.